Yihao Sun

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PROFESSIONAL PROFILE

A graduate student at Carnegie Mellon University with strong foundations in machine learning, multimodal AI, and AI infrastructure. Experienced in building, integrating, and evaluating end-to-end AI systems under real-world constraints, including edge deployment, simulation platforms, robotics, performance-critical training pipelines, and efficient vision-language/reasoning model research.

EDUCATIONAL BACKGROUND

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Mobile & Internet of Things Engineering — Anticipated May 2027
GPA: 3.80/4.0

University of Michigan — Ann Arbor, MI
Bachelor of Science in Engineering, Major in Data Science, Minor in Mathematics — Aug 2023 - May 2025
GPA: 3.86/4.0

Shanghai Jiao Tong University — Shanghai, China
Bachelor of Engineering, Major in Electrical and Computer Engineering, Minor in Computer Science — Sep 2021 - Aug 2025
GPA: 3.53/4.0

Honors & Awards: Dean's List (2023, 2024), University Honor (2023, 2024, 2025)
Relevant Coursework: Computer Vision, Conversational AI, Database Systems, Distributed Systems, Machine Learning Research Experience, Mobile and Pervasive Computing, Generative AI, Large Language Model Systems, Robotics Learning

PROFESSIONAL EXPERIENCE

AI Infrastructure Intern — May 2026 – Present
Baidu — Multimodal AI Research

- Investigating latent-space approaches for vision-reasoning models in an ongoing research project, emphasizing efficient multimodal reasoning, compact representations, and infrastructure-aware model design.
- Producing internal technical surveys on efficient vision-reasoning models, covering recent advances in multimodal reasoning, efficiency-aware post training, and inference-efficient systems architectures.
- Supported a related vision-language model research project on on-policy distillation by reviewing methods, assisting with experimental planning, and summarizing technical tradeoffs for internal research discussions.

RESEARCH EXPERIENCE & PUBLICATIONS

End-to-End Adaptive Smart Glasses for Caregiving — Oct 2025 - Present (Early-staged)
Carnegie Mellon University - Privacy-First Wearable AI Research,

- Developing a privacy-first wearable assistant using an edge Vision-Language Model (VLM) and Multi-Timescale Meta-Continual Learning framework that dynamically adapts sensor fusion between chronic monitoring (long-term health tracking) and acute injury modes (immediate emergency response).
- Implementing a Dual Working Memory system with Reinforcement Learning to personalize intervention timing, optimizing the complex tradeoff between assistance utility and user disruption—balancing real-time constraints with long-term adaptation through continuous learning.
- Designing the system architecture to operate entirely on edge hardware, ensuring data privacy while maintaining responsiveness for time-sensitive caregiving scenarios.

Part-Level Scene Understanding with LLMs May 2025 – Sept 2025
Open-Source Research Project — https://github.com/PSGBOT/KAFNet_Pipeline — IDETC/CIE 2026

- Constructed the comprehensive Part-Segmentation-Relation dataset through an automated pipeline that fuses Semantic-SAM for hierarchical segmentation with Gemini 2.5 for detailed kinematic annotation of part functionalities and relationships.
- Designed and implemented KAF-Net, a novel scene graph generation model that predicts fine-grained part relationships using Kinematic Affinity Fields and Multi-Channel Masks to capture both spatial and functional dependencies.
- Enabled kinematic-aware robotic task planning by integrating KAF-Net with Large Language Models to generate executable action sequences via dynamic graph pruning, integrating the SayPlan framework for practical robotic manipulation.
- **Publication Status:** *Co-first Author "KAF-Net: Towards Part-Informed Scene Graphs for Fine-Grained Robotic Manipulation"* .

TeraSim: Generative Simulation for Autonomous Vehicle Safety Mar 2024 – Apr 2025
MCITY Lab Research — https://arxiv.org/abs/2503.03629 — SAD Workshop at CVPR 2026

- Engineered a novel co-simulation plugin that integrates SUMO (traffic flow simulation) and CARLA (3D environment simulation) to inject dynamic roadblocks, vulnerable road users, and agent misfeasance into autonomous vehicle testing loops.
- Converted and refined OpenDrive maps for high-fidelity simulation of the MCITY test environment, improving map accuracy and providing critical feedback to the facility team for physical test track enhancements.
- **Publication Status:** Co-authored paper *"TeraSim: Uncovering Unknown Unsafe Events for Autonomous Vehicles through Generative Simulation"*.
- This research serves as the technical foundation for **SaferDrive.ai** (https://www.saferdrive.ai/), a startup focused on generative AI simulation for autonomous vehicle safety.

Magnetic Wristband Gesture Recognition Mar 2022 – Mar 2023
Shanghai Jiao Tong University Undergraduate Research

- Optimized hardware configuration and data processing pipelines to achieve >80% accuracy in full-finger gesture classification with high noise resistance in real-world environments.
- Developed noise-resistant signal processing techniques to mitigate external magnetic interference, implementing advanced filtering algorithms to isolate gesture signals from environmental noise.
- Mapped gestures via Euclidean space geometric center analysis, creating a robust classification system that maintains accuracy across different users and environmental conditions.

TECHNICAL SKILLS & PROFICIENCIES

Programming Languages: Python, C/C++, CUDA, SQL, R, Go, System Verilog, Java

AI & Machine Learning Tools: PyTorch, TensorFlowLite, Scikit-learn, NumPy, Pandas, Matplotlib, stable_baselines3, OpenCV, CARLA, SUMO, MuJoCo

Systems & Development Tools: Git, Docker, Linux/Unix, Web Systems, Databases

Computer Architecture & Hardware: RISC-V ISA, ARM ISA, Branch Prediction Algorithms, Superscalar CPU Pipelines

Concepts & Methodologies: Databases, Data Mining, Distributed Systems, Information Security, LLM Distillation, LLM Finetuning, Model Quantization, Reinforcement Learning, Robotics Learning, AI Infrastructure

Languages: English (Native/Bilingual Proficiency; TOEFL 112/120), Chinese (Native)

PROJECTS & IMPLEMENTATIONS

Humanoid Robot Backflip via Reinforcement Learning Jan 2026 – May 2026
Robotics Learning Course Project

- Developed a multi-stage training pipeline using PPO in MuJoCo Humanoid-v5, decomposing the backflip into sequential skills to avoid local optima inherent in end-to-end training of complex acrobatic motions.
- Applied skill transfer by initializing the backflip policy with pretrained high-jump network weights, enabling the agent to retain explosive vertical impulse while freely adapting to learn backward rotation and landing stabilization.
- Applied a gravity curriculum that progressively increased gravity from 0.3g to 1.0g during training, providing extended airtime in early stages for the agent to explore rotational behaviors that are unreachable through random exploration under nominal gravity.
- Designed phase-based dense reward functions with velocity-driven signals and angular velocity tracking, iteratively resolving reward hacking pathologies such as bouncing exploits and premature termination.

RetFormer: Retrieval-Augmented Document Understanding Jan 2026 – May 2026
Generative AI Course Project

- Designed and implemented RetFormer, a 23M-parameter learnable bridge between a frozen ColQwen2 retriever and a frozen Qwen-VL reader, compressing each retrieved document page into 32 evidence tokens via MaxSim-gated cross-attention to enable scalable multi-page document QA.
- Introduced a novel gating mechanism that biases cross-attention logits with per-patch retriever relevance scores, with a per-head per-layer learnable gate scalar; diagnosed a dead fixed-point failure at the original initialization and resolved it by rescaling $\tilde{\lambda}$ to $\text{softplus}^{-1}(5.0)$ paired with an auxiliary OCR loss, producing a graduated coarse-to-fine attention pattern across layers.
- Achieved Pareto-dominant performance over Top-1 raw page retrieval on MMLongBench-Doc (F1 0.241 vs. 0.213) at less than half the median reader token cost, and conducted systematic ablations falsifying the K-bottleneck hypothesis and characterizing out-of-distribution failure modes on chart imagery.
- Trained and evaluated across three retriever–reader stacks on NVIDIA H100 nodes, implementing vision token splicing with M-RoPE position encoding, a modality routing head with BM25 fallback, and a multi-page agentic inference loop.

Parallelism and Optimizations for Mamba-2 Models Jan 2026 – May 2026
Large Language Model System Course Project

- Designed and implemented the distributed parallelism framework for Mamba-2's Structured State Space Duality (SSD) algorithm, combining Megatron-style tensor parallelism for linear layers with sequence parallelism adapted to SSM recurrence via cross-device hidden state passing.
- Built communication primitives for chunked sequence parallelism that transmit only the compact, fixed-size SSM hidden state between GPUs, making inter-device communication cost independent of sequence length unlike Transformer KV-cache approaches.
- Implemented `ColumnParallelLinear` and `RowParallelLinear` distributed layers with proper all-reduce synchronization, enabling a 2×2 TP+SP grid that achieved up to $2.11\times$ speedup scaling to 1.38B parameters.
- Contributed to a system that, combined with IO-aware fused Triton kernels, reduced per-GPU memory from 31.3 GB to 1.1 GB and extended maximum sequence length from 16K to 262K tokens on a DNA sequence modeling task over the human reference genome.

Real-time Multiple Object Tracking on Edge Hardware Sept 2025 - Dec 2025
Mobile and Pervasive Computing Course Project

- Engineered a real-time Multiple Object Tracking pipeline on Google Pixel 7/9 Pro smartphones, utilizing on-device TPU acceleration for YOLO11 and ByteTrack for multi-object tracking.
- Implemented optical-flow-based prediction for intermediate frames to ensure stable and smooth 30 FPS tracking while maintaining accuracy.
- Developed techniques for keyframe ratio optimization and adaptive frame dropping to handle varying computational loads while preserving tracking continuity.

Impossible Distillation & LoRA Fine-Tuning Sept 2024 - Dec 2024
Machine Learning Research Experience Capstone Project

- Reproduced the "Impossible Distillation" framework to generate high-quality synthetic paraphrase datasets from GPT-2, implementing the complete dataset generation pipeline with rigorous semantic and syntactic evaluation metrics.
- Integrated Low-Rank Adaptation with the distillation pipeline, achieving a >99% reduction in trainable parameters while slightly improving model performance on paraphrasing tasks.
- Designed and implemented the parameter-efficient fine-tuning process, demonstrating practical approaches for adapting large language models with limited computational resources.
- Validated the pipeline quality through comprehensive evaluation including BLEU scores, semantic similarity metrics, and human evaluation studies.

3-Way Superscalar RISC-V Processor May 2025 - Aug 2025

Computer Architecture Capstone Project

- Designed, implemented, and optimized a high-performance, 3-way superscalar RISC-V processor from the ground up in System Verilog, implementing a 5-stage pipeline with comprehensive data and control hazard handling.
- Enhanced instruction throughput by designing an advanced TAGE branch predictor with aging useful bits mechanism, reducing misprediction rates to achieve a 30% performance uplift compared to baseline predictors.
- Developed extensive verification testbenches and performance analysis tools to validate correctness and measure architectural improvements under various workload conditions.

Distributed Sharded Key-Value Store Series Jan 2025 – Apr 2025

Distributed Systems Course Project — Go, Paxos, Distributed Consensus

- Architected a high-performance distributed storage system, evolving it from a Primary/Backup replication model managed by a central View Service to a scalable, Sharded Multi-Paxos architecture.
- Implemented a robust Paxos consensus library to manage replicated state machines, ensuring linearizable consistency and partition tolerance without relying on a central coordinator.
- Designed a ShardMaster configuration service to partition data across replica groups, employing a deterministic load-balancing algorithm to dynamically rebalance shards during cluster scaling with minimal data movement.
- Engineered fault-tolerance mechanisms including a View Service for automatic failover in the initial design, and later implemented log compaction and concurrent RPC handling to optimize the Paxos-based implementation.

Fake News Detection System Sep 2024 - Dec 2024

Conversational AI Course Project

- Architected a Retrieval-Augmented Generation based verification system that cross-references user queries against a curated database of credible sources, providing evidence-based fact-checking responses.
- Engineered a backend data pipeline to automatically crawl, process, and index news articles from reputable sources, enabling efficient, low-latency information retrieval with semantic search capabilities.
- Designed the system with scalability in mind, supporting real-time updates to the knowledge base as new information becomes available.

No-Limit Texas Hold'em AI Agent Jan 2026 - Present

Personal Research Project — https://github.com/Alexcel-Harry/poker_bot

- Developed a Deep Counterfactual Regret Minimization (Deep CFR) agent for multi-player No-Limit Hold'em, utilizing separate advantage and strategy networks to approximate Nash Equilibrium.
- Engineered a high-throughput training pipeline with 64-way parallelized environment traversals and batched FP16 inference, reducing convergence time to < 8 hours.
- Designing an action abstraction layer to map continuous betting dynamics into discrete strategy bins, overcoming the sparsity issues of No-Limit action spaces.